

R3

c - CDR, F - periodic downloaded static route, W - NHRP, 1 - LISP
a - application route
+ - replicated route, % - next hop override, p - overrides from PIR

Gateway of last resort is not set

S 10.0.0.0/8 [1/0] via 30.0.0.1
S 15.0.0.0/8 [1/0] via 172.19.0.2
S 20.0.0.0/8 [1/0] via 30.0.0.1
21.0.0.0/8 is variably subnetted, 2 subnets, 2 masks
C 21.0.0.0/8 is directly connected, Ethernet0/2
L 21.0.0.1/32 is directly connected, Ethernet0/2
30.0.0.0/8 is variably subnetted, 2 subnets, 2 masks
C 30.0.0.0/8 is directly connected, Ethernet0/0
L 30.0.0.2/32 is directly connected, Ethernet0/0
S 162.14.0.0/16 [1/0] via 21.0.0.2
172.19.0.0/16 is variably subnetted, 2 subnets, 2 masks
C 172.19.0.0/16 is directly connected, Ethernet0/1
L 172.19.0.1/32 is directly connected, Ethernet0/1
S 192.168.168.0/24 [1/0] via 21.0.0.2

R3#

R3#

R3#

R3#

E1 - ISPF external type 1, E2 - ISPF external type 2
 i - IS-IS, ex - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2
 ia - IS-IS inter area, * - candidate default, U - per-user static route
 o - ODR, F - periodic downloaded static route, H - NHRP, I - ISIS
 a - application route
 + - replicated route, % - next hop override, p - overrides from PfR

Gateway of last resort is not set

I

```

10.0.0.0/8 is variably subnetted, 2 subnets, 2 masks
C    10.0.0.0/8 is directly connected, Ethernet0/0
L    10.0.0.1/32 is directly connected, Ethernet0/0
S    15.0.0.0/8 [1/0] via 20.0.0.2
20.0.0.0/8 is variably subnetted, 2 subnets, 2 masks
C    20.0.0.0/8 is directly connected, Ethernet0/1
L    20.0.0.1/32 is directly connected, Ethernet0/1
S    21.0.0.0/8 [1/0] via 20.0.0.2
S    30.0.0.0/8 [1/0] via 20.0.0.2
S    162.14.0.0/16 [1/0] via 20.0.0.2
S    172.19.0.0/16 [1/0] via 20.0.0.2
S    192.168.168.0/24 [1/0] via 20.0.0.2
R1#
R1#
R1#
  
```

E1 - OSPF external type 1, E2 - OSPF external type 2
i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2
ia - IS-IS inter area, * - candidate default, U - per-user static route
o - ODR, P - periodic downloaded static route, H - NHRP, l - LISP
a - application route
+ - replicated route, % - next hop override, p - overrides from PfR

Gateway of last resort is not set

```
S 10.0.0.0/8 [1/0] via 20.0.0.1
S 15.0.0.0/8 [1/0] via 30.0.0.2
20.0.0.0/8 is variably subnetted, 2 subnets, 2 masks
C 20.0.0.0/8 is directly connected, Ethernet0/0
L 20.0.0.2/32 is directly connected, Ethernet0/0
S 21.0.0.0/8 [1/0] via 30.0.0.2
30.0.0.0/8 is variably subnetted, 2 subnets, 2 masks
C 30.0.0.0/8 is directly connected, Ethernet0/1
L 30.0.0.1/32 is directly connected, Ethernet0/1
S 162.14.0.0/16 [1/0] via 30.0.0.2
S 172.19.0.0/16 [1/0] via 30.0.0.2
S 192.168.168.0/24 [1/0] via 30.0.0.2
```

R2#

R2#

R2#

72.19.0.2

e0/1

e0/0

21.0.0.2

R5

Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
E1 - OSPF external type 1, E2 - OSPF external type 2
i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2
ia - IS-IS inter area, * - candidate default, U - per-user static route
o - ODR, P - periodic downloaded static route, H - NHRP, I - IIS
a - application route
+ - replicated route, % - next hop override, p - overrides from PfR

Gateway of last resort is not set

```
S 10.0.0.0/8 [1/0] via 21.0.0.1
S 15.0.0.0/8 [1/0] via 21.0.0.1
S 20.0.0.0/8 [1/0] via 21.0.0.1
S 21.0.0.0/8 is variably subnetted, 2 subnets, 2 masks
C   21.0.0.0/8 is directly connected, Ethernet0/0
L   21.0.0.2/32 is directly connected, Ethernet0/0
S 30.0.0.0/8 [1/0] via 21.0.0.1
S 162.14.0.0/16 [1/0] via 192.168.168.2
S 172.19.0.0/16 [1/0] via 21.0.0.1
S 192.168.168.0/24 is variably subnetted, 2 subnets, 2 masks
C   192.168.168.0/24 is directly connected, Ethernet0/1
L   192.168.168.1/32 is directly connected, Ethernet0/1
```

R5#



E1 - OSPF external type 1, E2 - OSPF external type 2
i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2
ia - IS-IS inter area, * - candidate default, U - per-user static route
o - ODR, P - periodic downloaded static route, H - NHRP, l - LISP
a - application route
+ - replicated route, % - next hop override, p - overrides from PfR

Gateway of last resort is not set

```
S    10.0.0.0/8 [1/0] via 172.19.0.1
    15.0.0.0/8 is variably subnetted, 2 subnets, 2 masks
C    15.0.0.0/8 is directly connected, Ethernet0/0
L    15.0.0.1/32 is directly connected, Ethernet0/0
S    20.0.0.0/8 [1/0] via 172.19.0.1
S    21.0.0.0/8 [1/0] via 172.19.0.1
S    30.0.0.0/8 [1/0] via 172.19.0.1
S    162.14.0.0/16 [1/0] via 172.19.0.1
    172.19.0.0/16 is variably subnetted, 2 subnets, 2 masks
C    172.19.0.0/16 is directly connected, Ethernet0/1
L    172.19.0.2/32 is directly connected, Ethernet0/1
S    192.168.168.0/24 [1/0] via 172.19.0.1
```

R4#

R4#

R4#

172.1
21.0

R6

N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
E1 - OSPF external type 1, E2 - OSPF external type 2
i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2
ia - IS-IS inter area, * - candidate default, U - per-user static route
o - ODR, P - periodic downloaded static route, H - NHRP, l - LISP
a - application route
+ - replicated route, % - next hop override, p - overrides from Pfr

Gateway of last resort is not set

S 10.0.0.0/8 [1/0] via 192.168.168.1
S 15.0.0.0/8 [1/0] via 192.168.168.1
S 20.0.0.0/8 [1/0] via 192.168.168.1
S 21.0.0.0/8 [1/0] via 192.168.168.1
S 30.0.0.0/8 [1/0] via 192.168.168.1
162.14.0.0/16 is variably subnetted, 2 subnets, 2 masks
C 162.14.0.0/16 is directly connected, Ethernet0/0
L 162.14.0.1/32 is directly connected, Ethernet0/0
S 172.19.0.0/16 [1/0] via 192.168.168.1
192.168.168.0/24 is variably subnetted, 2 subnets, 2 masks
C 192.168.168.0/24 is directly connected, Ethernet0/1
L 192.168.168.2/32 is directly connected, Ethernet0/1

R6#

R6#

172.19.0.2

21.0.0.2

C:\WINDOWS\system32\cmd.exe

Windows IP Configuration

Ethernet adapter Local Area Connection:

Connection-specific DNS Suffix . :
IP Address. : 10.0.0.10
Subnet Mask : 255.0.0.0
Default Gateway : 10.0.0.1

C:\Documents and Settings\user>ping 15.0.0.10

Pinging 15.0.0.10 with 32 bytes of data:

Reply from 15.0.0.10: bytes=32 time=3ms TTL=124
Reply from 15.0.0.10: bytes=32 time=1ms TTL=124
Reply from 15.0.0.10: bytes=32 time=2ms TTL=124
Reply from 15.0.0.10: bytes=32 time=2ms TTL=124

Ping statistics for 15.0.0.10:

Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
Minimum = 1ms, Maximum = 3ms, Average = 2ms

C:\Documents and Settings\user>

```
C:\WINDOWS\system32\cmd.exe
Reply from 15.0.0.10: bytes=32 time=3ms TTL=124
Reply from 15.0.0.10: bytes=32 time=1ms TTL=124
Reply from 15.0.0.10: bytes=32 time=2ms TTL=124
Reply from 15.0.0.10: bytes=32 time=2ms TTL=124

Ping statistics for 15.0.0.10:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 1ms, Maximum = 3ms, Average = 2ms

C:\Documents and Settings\user>ping 162.14.0.10

Pinging 162.14.0.10 with 32 bytes of data:

Reply from 162.14.0.10: bytes=32 time=4ms TTL=123
Reply from 162.14.0.10: bytes=32 time=3ms TTL=123
Reply from 162.14.0.10: bytes=32 time=2ms TTL=123
Reply from 162.14.0.10: bytes=32 time=2ms TTL=123

Ping statistics for 162.14.0.10:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 2ms, Maximum = 4ms, Average = 2ms

C:\Documents and Settings\user>
```


C:\WINDOWS\system32\cmd.exe

Windows IP Configuration

Ethernet adapter Local Area Connection:

Connection-specific DNS Suffix	:	:
IP Address	:	15.0.0.10
Subnet Mask	:	255.0.0.0
Default Gateway	:	15.0.0.1

C:\Documents and Settings\user>ping 10.0.0.1

Pinging 10.0.0.10 with 32 bytes of data:

Reply from 10.0.0.10: bytes=32 time=2ms TTL=124
Reply from 10.0.0.10: bytes=32 time=2ms TTL=124
Reply from 10.0.0.10: bytes=32 time=3ms TTL=124
Reply from 10.0.0.10: bytes=32 time=2ms TTL=124

Ping statistics for 10.0.0.10:

Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
Minimum = 2ms, Maximum = 3ms, Average = 2ms

C:\Documents and Settings\user>

C:\WINDOWS\system32\cmd.exe

Reply from 10.0.0.10: bytes=32 time=2ms TTL=124
Reply from 10.0.0.10: bytes=32 time=2ms TTL=124
Reply from 10.0.0.10: bytes=32 time=3ms TTL=124
Reply from 10.0.0.10: bytes=32 time=2ms TTL=124

Ping statistics for 10.0.0.10:

Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
Minimum = 2ms, Maximum = 3ms, Average = 2ms

C:\Documents and Settings\user>ping 162.14.0.10

Pinging 162.14.0.10 with 32 bytes of data:

Reply from 162.14.0.10: bytes=32 time=2ms TTL=124
Reply from 162.14.0.10: bytes=32 time=3ms TTL=124
Reply from 162.14.0.10: bytes=32 time=2ms TTL=124
Reply from 162.14.0.10: bytes=32 time=2ms TTL=124

Ping statistics for 162.14.0.10:

Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
Minimum = 2ms, Maximum = 3ms, Average = 2ms

C:\Documents and Settings\user>

C:\WINDOWS\system32\cmd.exe

Windows IP Configuration

Ethernet adapter Local Area Connection:

Connection-specific DNS Suffix . :
IP Address : 162.14.0.10
Subnet Mask : 255.255.0.0
Default Gateway : 162.14.0.1

C:\Documents and Settings\user>ping 10.0.0.10

Pinging 10.0.0.10 with 32 bytes of data:

Reply from 10.0.0.10: bytes=32 time=2ms TTL=123
Reply from 10.0.0.10: bytes=32 time=3ms TTL=123
Reply from 10.0.0.10: bytes=32 time=2ms TTL=123
Reply from 10.0.0.10: bytes=32 time=2ms TTL=123

Ping statistics for 10.0.0.10:

Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
Minimum = 2ms, Maximum = 3ms, Average = 2ms

C:\Documents and Settings\user>

My Computer

My Network Places

Recycle Bin

Internet Explorer

Cisco ASDM-1...

Java ASDM

PUTTY

C:\WINDOWS\system32\cmd.exe

Reply from 10.0.0.10: bytes=32 time=2ms TTL=123
Reply from 10.0.0.10: bytes=32 time=3ms TTL=123
Reply from 10.0.0.10: bytes=32 time=2ms TTL=123
Reply from 10.0.0.10: bytes=32 time=2ms TTL=123

Ping statistics for 10.0.0.10:

Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
Minimum = 2ms, Maximum = 3ms, Average = 2ms

C:\Documents and Settings\user>ping 15.0.0.10

Pinging 15.0.0.10 with 32 bytes of data:

Reply from 15.0.0.10: bytes=32 time=3ms TTL=124
Reply from 15.0.0.10: bytes=32 time=3ms TTL=124
Reply from 15.0.0.10: bytes=32 time=3ms TTL=124
Reply from 15.0.0.10: bytes=32 time=2ms TTL=124

Ping statistics for 15.0.0.10:

Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
Minimum = 2ms, Maximum = 3ms, Average = 2ms

C:\Documents and Settings\user>